ModelIt!

Teacher's Guide

Why Hurricanes Are Getting Angrier

The Science Behind Supercharged Storms

Quick Reference

Grade Level	Grade 8
NGSS Standard	MS-ESS2-5, MS-ESS2-6
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: MS-ESS2-5, MS-ESS2-6

Collect data to provide evidence for how the motions and complex interactions of air masses results in changes in weather conditions. Develop and use a model to describe how unequal heating of the Earth causes patterns of atmospheric and oceanic circulation that determine regional climates.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model showing how ocean temperature and atmospheric conditions determine hurricane intensity.
Disciplinary Core Idea	**ESS2.D Weather and Climate** — Complex interactions of air masses, ocean temperatures, and atmospheric conditions drive weather patterns.
Crosscutting Concept	**Energy and Matter** — Students trace the flow of thermal energy from ocean water through evaporation into the kinetic energy of hurricane winds.

CAST Testing Alignment

MULTIPLE CHOICE:

Sea surface temperatures in the Gulf of Mexico rise from 82°F to 88°F. A tropical storm enters this region. What is the most likely result?

CONSTRUCTED RESPONSE:

Using your model, explain the chain of events from warm ocean water to destructive storm surge in a coastal city.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Hurricanes are the most powerful weather events on Earth. A single hurricane can release energy equivalent to 10,000 nuclear bombs. Understanding the science behind their formation and intensification is critical as climate change alters ocean temperatures.

Hurricane Formation

Hurricanes form over warm ocean water (above 80°F/26.5°C). Warm water evaporates, and the moist air rises rapidly through convection. As it rises, it cools and condenses, releasing latent heat that further fuels the rising air. The Coriolis effect from Earth's rotation causes the system to spin.

The Energy Connection

A hurricane is essentially a heat engine. The warmer the ocean surface, the more water evaporates, and the more energy is available to fuel the storm. The relationship between sea surface temperature and maximum potential hurricane intensity is well-established in atmospheric science.

Rapid Intensification

Rapid intensification occurs when a hurricane's maximum sustained winds increase by 35+ mph in 24 hours. Hurricane Patricia (2015) went from tropical storm to Category 5 in just 24 hours. Studies show RI events have increased in frequency as oceans warm, making hurricanes harder to forecast and evacuate for.

Climate Change Connection

Global average sea surface temperatures have risen about 1.5°F since 1900. The proportion of Category 4 and 5 hurricanes has increased significantly since 1980. Climate models predict continued intensification as oceans absorb more heat, with the strongest storms getting even stronger.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify sea surface temperature, atmospheric moisture, hurricane wind speed, and storm surge height as system components.

E **Establish Relationships**
Students determine that warmer water and more moisture both increase wind speed (positive), and wind speed increases storm surge (positive).

V **Visualize & Model**
Students build the hurricane model showing how ocean heat translates to storm intensity.

E **Evaluate Outcomes**
Students compare normal, supercharged, and cool water scenarios to see the energy relationship.

R **Revise & Extend**
Students add wind shear or coastal population to explore additional factors.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"This storm has the energy of 10,000 nuclear bombs. Where does all that power come from?"*

Watch For

Show dramatic satellite footage. Let the scale sink in.

Slide 2: Learning Objectives

3 min

Say: *"Today we model the engine that drives the most powerful storms on Earth."*

Watch For

Discuss recent hurricanes students may remember.

Slide 3: Big Question

3 min

Say: *"Hurricane records show storms are intensifying faster than ever. Why?"*

Watch For

Show before/after photos of hurricane damage.

Slide 4: LEVER Framework

2 min

Say: *"We'll build a model to understand what fuels these monster storms."*

Watch For

Preview the modeling process.

Slide 5: Activity 1: Components

8 min

Say: *"What inputs fuel a hurricane? What outputs do we measure?"*

Watch For

Guide sorting. Emphasize the ocean as the energy source.

Slide 6: Activity 2: Connections

8 min

Say: *"Warm water is the FUEL. More fuel = more powerful engine."*

Watch For

Map the energy pathway from ocean to wind to surge.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's see what a few degrees of ocean warming does to storm strength."*

Watch For

Students predict, then run. The supercharged scenario is dramatic.

Slide 8: Discoveries

5 min

Say: *"Just 6°F of warmer water turned a Category 1 into a Category 5."*

Watch For

Connect to real hurricane data and climate trends.

Slide 9: STEM Challenge

12 min

Say: *"A coastal city was devastated. They hired YOU to redesign the community."*

Watch For

Teams design hurricane-resilient communities.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Sea Surface Temperature, Atmospheric Moisture
- **INTERNAL:** Hurricane Wind Speed, Storm Surge Height

Relationships

- **Sea Surface Temperature to Hurricane Wind Speed:** POSITIVE (+) — Warmer water evaporates more, providing more energy through latent heat to fuel stronger winds.
- **Atmospheric Moisture to Hurricane Wind Speed:** POSITIVE (+) — More moisture means more condensation, which releases more latent heat to power the storm.
- **Hurricane Wind Speed to Storm Surge Height:** POSITIVE (+) — Stronger winds push more ocean water onto shore, creating higher and more destructive storm surges.

Key Discoveries

- Hurricanes are heat engines — warm ocean water is their fuel
- Just a few degrees of extra ocean warmth dramatically increases storm intensity
- Rapid intensification is becoming more common as oceans warm
- When hurricanes hit cool water or land, they lose their energy source and weaken

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Hurricanes are getting more frequent"	The total NUMBER of hurricanes hasn't clearly increased, but the PROPORTION of intense storms (Category 4-5) has increased significantly. Fewer storms, but the ones that form are stronger.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Wind is the most dangerous part"	Storm surge actually causes the most deaths and property damage during hurricanes. A 15-foot wall of ocean water pushed inland is more destructive than wind alone.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Climate change creates hurricanes"	Climate change doesn't CREATE hurricanes — they've always existed. But warmer oceans provide more FUEL for storms that do form, making them stronger and more likely to rapidly intensify.